

MINI PROJECT

Aim: To create a functional website with frontend and backend

Abstract:

VistaVignettes is a modern, responsive web application developed as a high-resolution wallpaper gallery. The project successfully integrates cutting-edge web technologies with artificial intelligence to deliver a dynamic user experience. The primary goal was to automate content management and enhance user discovery, which was achieved by employing the Gemini API for intelligent, multi-modal analysis of images, automated metadata generation, and tagging. The frontend features an immersive Three.js 3D interactive sphere for showcasing content and a fluid Masonry gallery for browsing. This project demonstrates proficiency in full-stack web development (PHP/MySQL), client-side 3D rendering, and integration of generative AI services, resulting in a robust, searchable, and dynamically curated content library.

Introduction:

The consumption of digital content, particularly high-quality visual media, necessitates robust platforms for efficient organization and discovery. This project, VistaVignettes, addresses this need by creating a technologically advanced wallpaper gallery.

The application leverages the power of modern web technologies, including:

Immersive Visualization: Three.js is used to introduce a unique 3D interactive element, moving beyond static web design.

Intelligent Content Curation: The Gemini API is integrated into the content pipeline to streamline the typically manual process of tagging and cataloging visual assets.

This report details the design, development, and implementation of this dynamic platform, emphasizing the core role of the AI integration in establishing an intelligent content architecture.

Methodology:

The development of VistaVignettes followed a structured phase-based methodology:

1. Planning & Design

Defined the responsive layout (Masonry), Glassmorphism theme, and database schema.

Identified the need for a non-traditional visual component (3D Sphere).

Database structure (images table), Wireframes.

2. Frontend Development

Built the core HTML structure (index.html, gallery.html). Implemented responsive CSS using a Glassmorphism aesthetic (css/style.css).

Responsive HTML/CSS structure.

3. Three.js 3D Implementation

Integrated the Three.js library and developed JavaScript logic (js/main.js) to render and control the interactive 3D sphere, mapping featured wallpapers onto its surface.

Functional 3D visualization on the homepage.

4. Backend Development (PHP/MySQL)

Developed the MySQL database (vista_db) and the fundamental PHP API scripts (api/get_images.php) for data retrieval and filtering.

Working database and data access API.

5. AI Integration & Seeding

Implemented the core logic in api/seed_images.php to interface with the Gemini API. Configured the script to perform multi-modal analysis and return structured JSON for database insertion.

Automated content metadata generation.

- **Database Schema**

The core database is vista_db with a single table, images:

```
CREATE TABLE images (  
  id INT(11) AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(255) NOT NULL,  
  description TEXT, /* AI-generated metadata */  
  filepath VARCHAR(255) NOT NULL,  
  tags VARCHAR(255), /* AI-generated, comma-separated tags */  
  is_sphere_image BOOLEAN DEFAULT 0 /* Content separation flag */  
);
```

- **Gemini API Functionality (api/seed_images.php)**

This script is the central integration point. It reads a new image file, encodes it, sends it to the Gemini API, and uses the returned generated text (containing description and tags) to execute a robust INSERT query into the images table.

```
// Pseudo-code for image analysis and database update  
$image_data = file_get_contents($image_path);
```

```

$base64_image = base64_encode($image_data);

// Call to Gemini API for multi-modal analysis
$gemini_response = call_gemini_api($base64_image, "Generate a concise description
and 5 relevant comma-separated tags for this image.");

// Extract and sanitize AI-generated metadata (description and tags)
$new_description = $gemini_response['description'];
$new_tags = $gemini_response['tags'];

// Database insertion logic using prepared statements
$sql = "INSERT INTO images (name, description, filepath, tags, is_sphere_image)
VALUES (?, ?, ?, ?, ?)";
// ... execute SQL

```

Sourcecode:

Index.html:

```

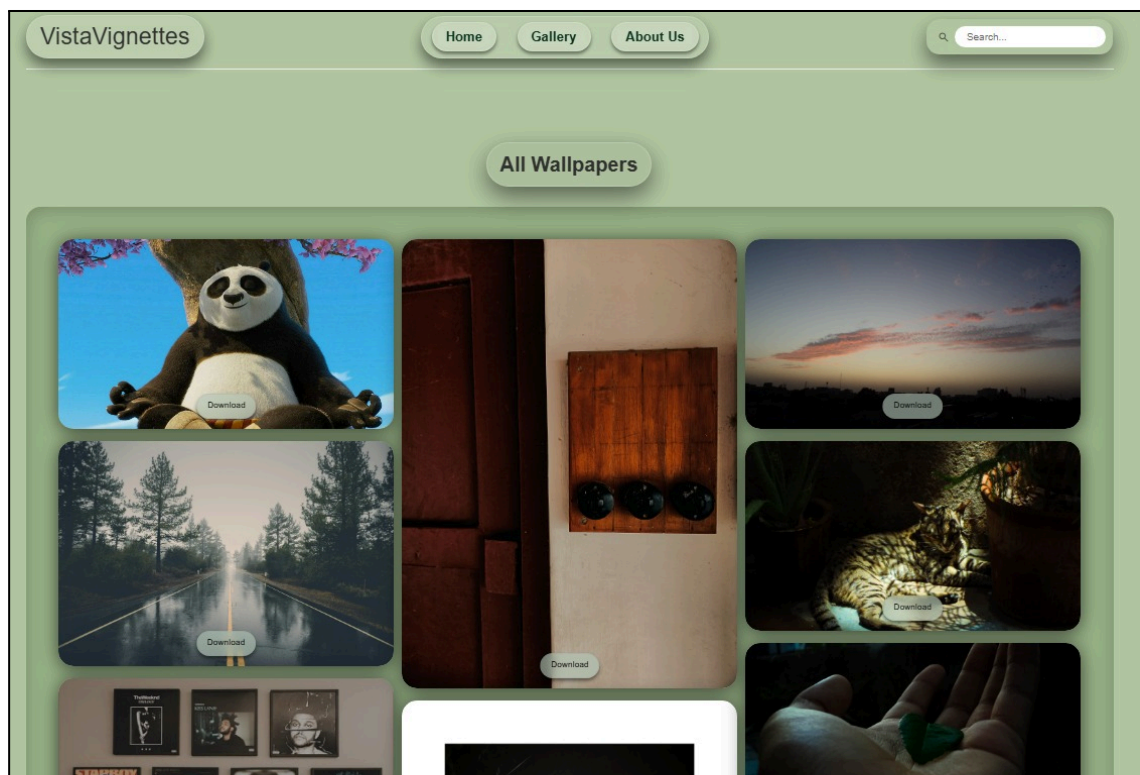
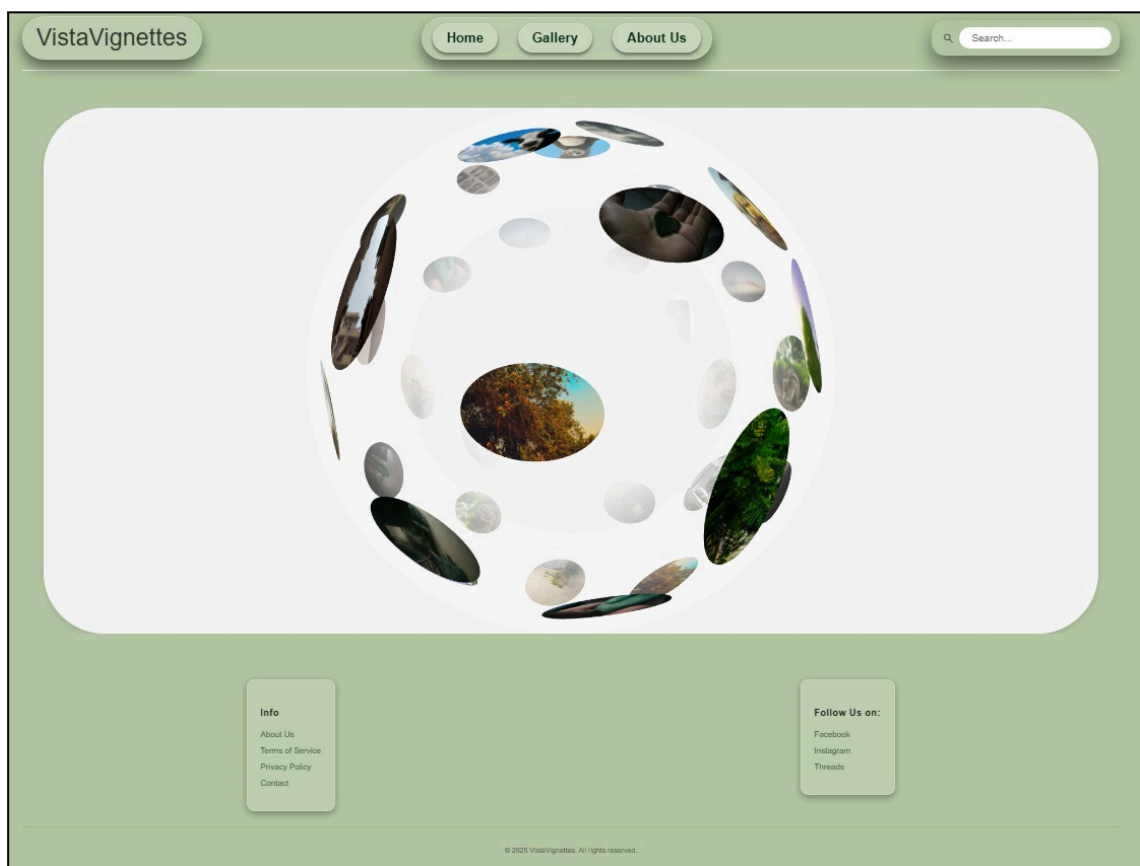
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>VistaVignettes</title>
  <link rel="stylesheet" href="css/style.css">
  <link rel="stylesheet"
href="https://fonts.googleapis.com/css2?family=Material+Symbols+Outlined:opsz,
wght,FILL,GRAD@20..48,100..700,0..1,-50..200" />
  <link rel="stylesheet"
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800;9
00&display=swap">
</head>
<body>
  <div class="container">
    <header>
      <div class="logo">
        <h1><a href="index.html">VistaVignettes</a></h1>
      </div>
      <div class="navigation-bar">
        <nav>
          <ul>
            <li><a href="index.html">Home</a></li>
            <li><a href="gallery.html">Gallery</a></li>
            <li><a href="about.html">About Us</a></li>

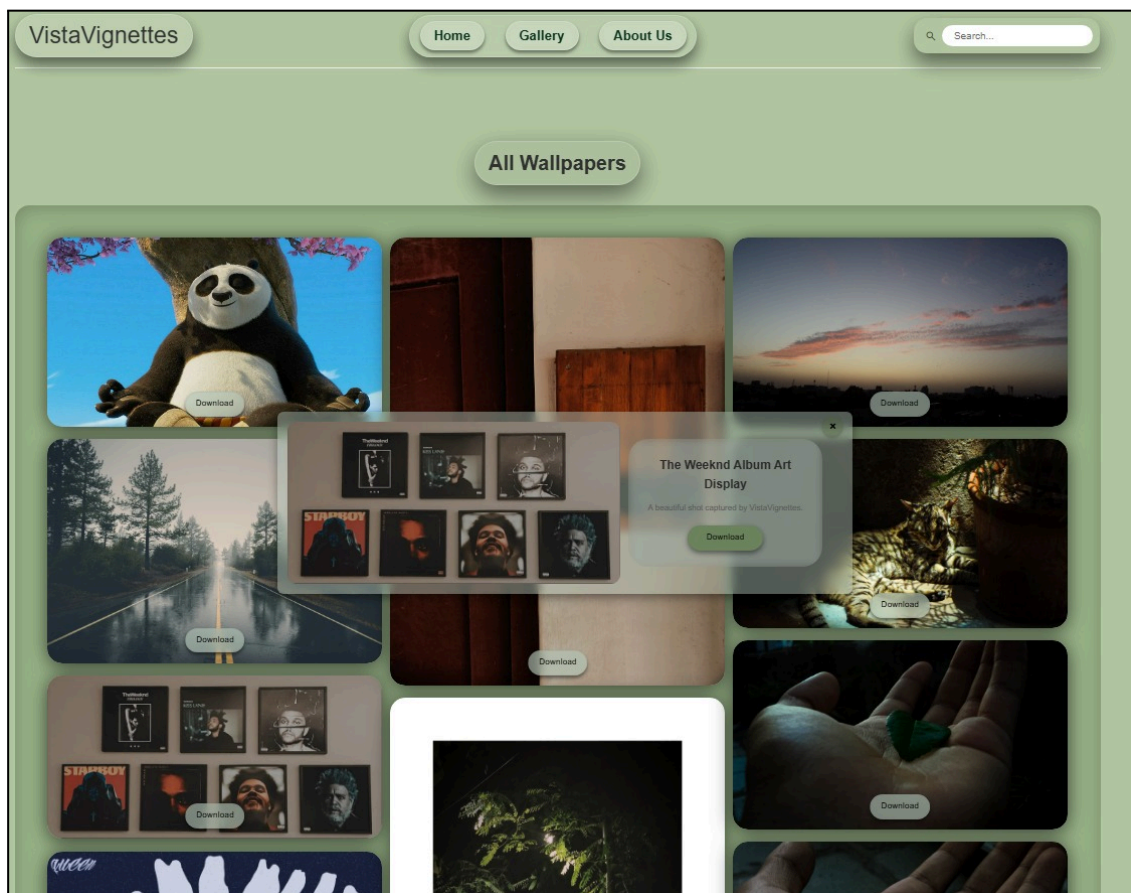
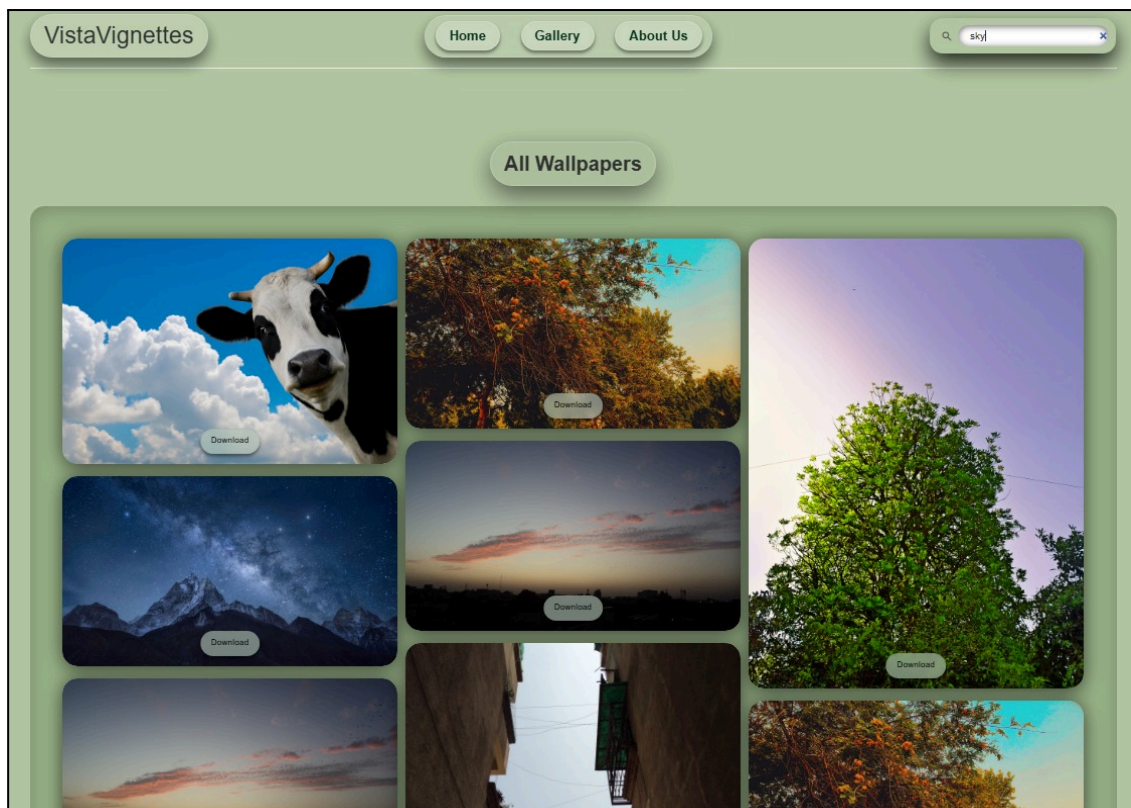
```

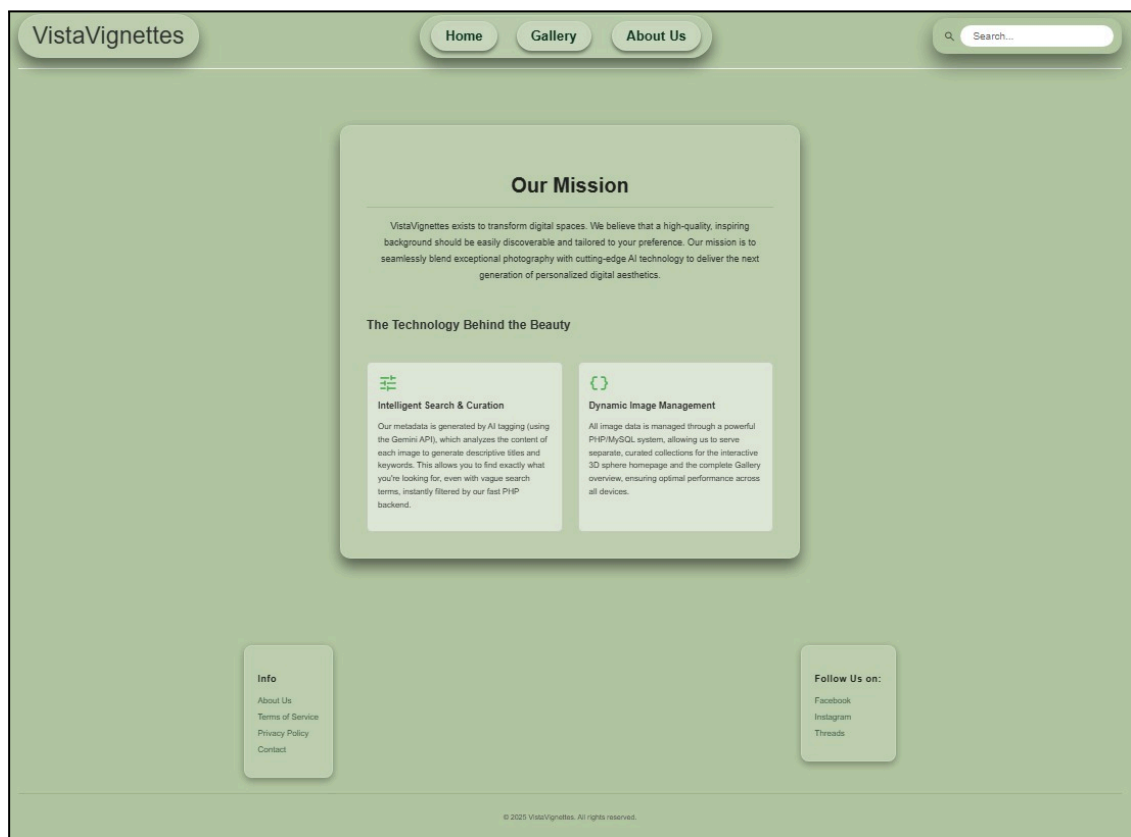
```

        </ul>
    </nav>
</div>
<form class="search-bar" action="gallery.html" method="GET">
    <button type="submit" class="search-icon-button" aria-label="Search">
        <span class="search-icon material-symbols-outlined">search</span>
    </button>
    <input class="search-input" name="search_term" type="search"
placeholder="Search...">
</form>
</header>
<main>
    <canvas id="threeJsCanvas"></canvas>
</main>
<footer>
    <div class="footer-section">
        <h5>Info</h5>
        <ul>
            <li><a href="about.html">About Us</a></li>
            <li><a href="#">Terms of Service</a></li>
            <li><a href="#">Privacy Policy</a></li>
            <li><a href="contact.html">Contact</a></li>
        </ul>
    </div>
    <div class="footer-section">
        <h5>Follow Us on:</h5>
        <ul>
            <li><a href="https://www.facebook.com/"
target="_blank">Facebook</a></li>
            <li><a
href="https://www.instagram.com/vistavignettes_in?utm_source=ig_web_button_sh
are_sheet&igsh=MXVxeWZqNzdvdvNHQ3cA=="
target="_blank">Instagram</a></li>
            <li><a href="https://threads.net/" target="_blank">Threads</a></li>
        </ul>
    </div>
    <div class="copyright">
        <p>&copy; 2025 VistaVignettes. All rights reserved.</p>
    </div>
</footer>
</div>
<script type="module" src="js/main.js"></script>
</body>
</html>

```







Result:

The VistaVignettes project successfully achieved its objectives of creating a visually engaging and intelligently managed content platform.

Successful AI Integration: The implementation of the Gemini API within the PHP pipeline for automated metadata generation was a significant achievement, proving the concept of AI-driven content seeding.

Enhanced UX: The Three.js 3D sphere provides a distinctive and modern user experience.

Full-Stack Proficiency: The project required the integration of multiple technologies—frontend rendering, PHP backend, MySQL database, and external AI services—demonstrating comprehensive web technology skills.

This project provided invaluable hands-on experience in building scalable, modern web solutions, emphasizing the future trend of integrating generative AI capabilities into standard web architecture.

Conclusion:

VistaVignettes successfully delivers a modern, user-friendly platform for browsing and discovering high-resolution wallpapers. The combination of a clean, responsive design, immersive 3D interaction using Three.js, and automated content management tools results in a polished and engaging experience.

By streamlining the process of uploading and organizing images, especially through features like automatic tagging and descriptions, the platform reduces manual workload while maintaining a well-structured and searchable gallery. The use of PHP and MySQL on the backend, along with JavaScript on the frontend, highlights a strong understanding of full-stack development and web integration.

Overall, this project demonstrates practical skills in building interactive, visually rich web applications and sets the stage for future improvements such as user accounts, favorites, or more advanced filtering options. VistaVignettes is not just a gallery, it's a solid foundation for a creative and evolving visual content platform.